Abagnale:

Improved thinking through dialogue

Abagnale is a physical, voice-based device that helps designers capture ideas and receive persona-based feedback during the design process.

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Abstract

This project explores how AI can be leveraged for novel interactive experiences that were previously possible only with human collaborators. Using IDEO and Delft design methods, I conducted a conceptual exploration to understand AI's capabilities in mediating design interactions. The work highlights how AI can act as a personalized, low-risk partner for idea capture and feedback, revealing both the opportunities for new value and the tensions that arise when AI misinterprets context. This pictorial demonstrates a systematic approach for integrating emerging technologies into design without forcing interaction, emphasizing experience over functionality.

Keywords

Human–AI Interaction, Exploratory Design, Interaction Design, Physical Computing, Design Ideation, Design Thinking

Introduction

Artificial Intelligence holds possibilities that designers do not yet fully understand, leaving them without the knowledge needed to design effectively with it. This project investigates AI's strengths and limitations through explorative design, asking:

> ***How might we design AI-enabled interactions as probes to discover what AI can do?***

The goal is to help users gain insight and reflect on AI's role in products, with the hope that designers can thoughtfully embed AI in meaningful experiences. The end result is a high-fidelity prototype of an interaction that surfaces these insights.

To approach this, I employed a systematic design process combining IDEO and TU Delft design thinking methodologies with an iterative design approach. The process is presented in five parts:

1. Initial research on AI capabilities
2. Explorative diverging
3. Concepting promising ideas
4. Evaluating and committing to a concept
5. Designing the final concept

Through this structured exploration, the work surfaces both opportunities and tensions in AI-mediated interactions, providing insights into how designers can shape emerging technologies into valuable, reflective experiences.

Phase I: Initial Research on AI Capabilities

The first phase focuses on understanding AI as a design material. This phase aims to characterize its fundamental properties, behaviors, and constraints.

In any design process, outcomes are limited by how well the designer understands the material they are working with. Because AI behaves differently from traditional computational systems, existing interaction paradigms do not fully apply. To design meaningful AI-enabled interactions, designers must first understand what AI is good at, what it struggles with, and how it differs from both humans and conventional software. This phase establishes that understanding, enabling more intentional and grounded explorative divergence later in the process.

Unlike traditional software, AI systems are probabilistic rather than deterministic. They do not follow instructions in a fixed way; instead, they interpret intent and generate the most likely response based on training data (Feng et al., 2023; Bender et al., 2021; Tribe AI, n.d.).

Strengths:

1. Operate at scale and speed, analyzing large datasets efficiently (Tribe AI, n.d.).
2. Recognize complex patterns beyond human capability (Tribe AI, n.d.).
3. Adapt and personalize responses over time (Dove et al., 2017; Shi et al., 2023).
4. Generate and synthesize new content rather than retrieve fixed outputs (Shi et al., 2023).
5. Reason and predict within bounded contexts (Shi et al., 2023).
6. Understand human language, enabling natural communication through text and speech (Feng et al., 2023).

These capabilities differentiate AI from traditional rule-based systems and open new possibilities for interaction.

Limitations and risks

1. Lack of true contextual understanding — AI does not "see the bigger picture"
2. Unreliable interpretation of intent, leading to misunderstandings
3. Risk of Bias (Shi et al., 2023).

AI is often described as "human-like," yet key differences remain:

Humans	AI
Contextual and situational	Probabilistic and statistical
Navigate ambiguity intuitively	Process large datasets
Slow but reflective	Fast but unreliable
True understanding	Approximate interpretation

Because AI is non-deterministic and operates with a reduced locus of control, it behaves less like a passive tool and more like an autonomous agent. When allowed to say "no," reinterpret input, or act independently, AI is no longer a "dumb" product. Instead, it introduces a new dimension of interaction where the system actively participates in shaping outcomes.

This ability to "think" independently marks a shift in human–computer interaction, enabling experiences previously only possible with other humans.

Defining Interaction

For this project, interaction is considered as a dialogue between two entities. Interaction does not require language (e.g., touching a soft versus sharp surface).

What makes AI interaction unique is that both parties are active agents. Unlike static products, AI engages in a dynamic, collaborative exchange with the user.

Text and speech currently dominate AI interaction because they mirror how humans are used to communicating with other intelligent entities (other humans). However, this also reveals an opportunity:

If AI is truly an interactive agent, what other forms of interaction are possible? Beyond speech, AI could engage through physiological signals, gestures, muscle movement, or other embodied inputs.

Phase II: Explorative Diverging

This phase explores a wide range of possible AI-enabled interactions to identify design probes that can surface how users interact with, interpret, and respond to AI. Rather than aiming for desirebility or marketfit, the focus is on generating concepts that deliberately expose tensions in human–AI interaction.

Using the project framing, *How might we design AI-enabled interactions as probes to discover what AI can do?*, a series of "How Might We" prompts were created to guide ideation. This resulted in approximately 40 initial concepts,

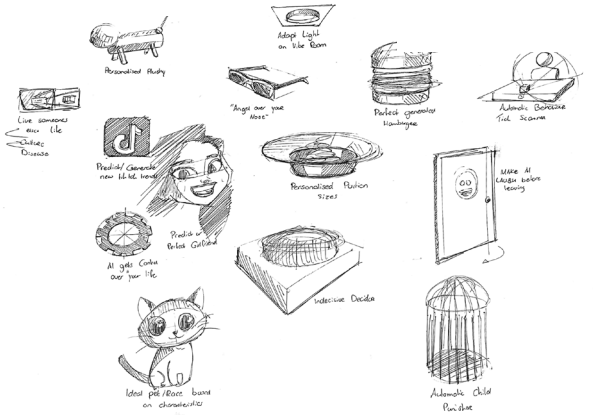
Rather than selecting ideas immediately, an incubation period was used. The concepts were left to rest while discussions took place with the course instructor, fellow designers, and AI experts (NVIDIA and AI students). This process allowed ideas to evolve, recombine, and sharpen beyond their initial forms before any evaluation occurred. Two ideas are chosen to further develop:

Concept 1 - Experiencing another lived reality

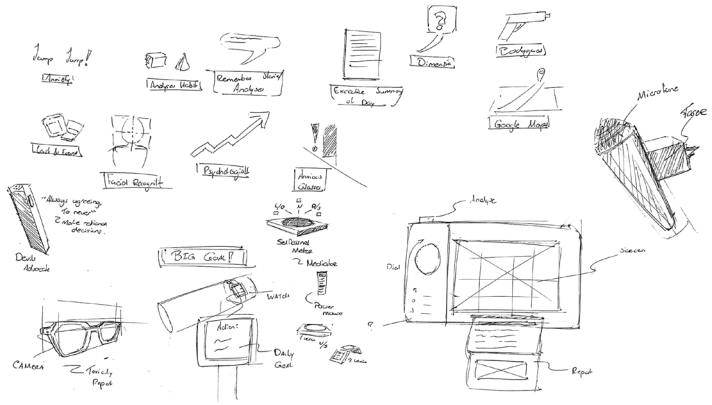
This concept explores glasses that analyze the user's surroundings through the perspective of a person with anxiety, overlaying interpreted thoughts and reactions onto a heads-up display. The concept addresses a recurring design challenge: the difficulty for designers to truly understand the lived experience of the people they design for. The main question this tries to answer is: "How might designers step into another person's lived reality through AI-mediated interpretation?"

Concept 2 - The disagreeable AI

This concept intentionally inverts common AI behavior by creating a probe whose goal is to resist agreement. Instead of supporting decisions, the AI challenges assumptions, introduces doubt, and complicates choices. The intent is to explore whether friction and disagreement can lead to more grounded, reflective decision-making by users. Inspired by my own difficulty choosing a concept for this course.



Question: How might we design a product to explore AI interactions?



Question: How migt we utalize on AI that senses surroundings?

Phase III: Concepting Promising Ideas

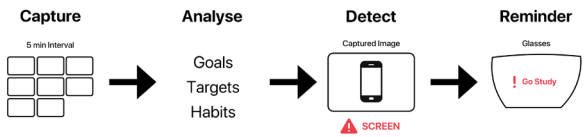
This phase further develops the most promising concepts into coherent interaction probes and evaluates them to determine which should be realized as a high-fidelity prototype. Both concepts were further iterated through sketching, system reasoning, and scenario development.

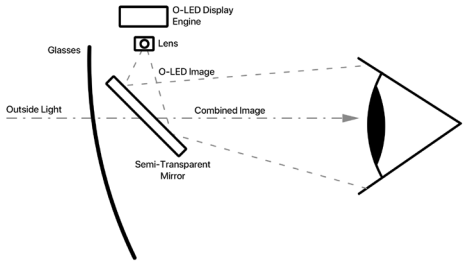
Concept 1 - Angel of your Shoulder Glasses

This concept evolved into a system where smart glasses periodically capture images of the user's environment and analyze long-term behavior patterns. When a recurring negative behavior is detected, the system subtly intervenes by displaying prompts intended to redirect the user's actions.

Design Question:

What if an AI understood your behavior better than you do, and quietly steered you toward the person it believes you should be?



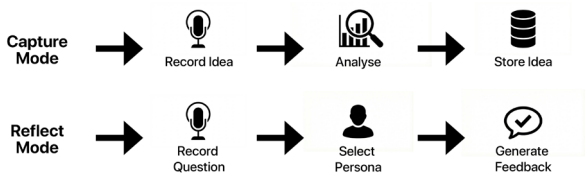


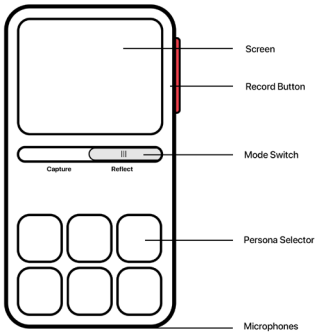
Concept 2 - Improved thinking through dialogue

The second concept reframes AI as a reflective partner in the design process. A voice-based device captures designers' ideas and reflects them back through distinct personas (e.g., environmental, business, user). By responding from these unique viewpoints, the AI helps designers quickly identify blind spots and make more informed, balanced decisions.

Design Question:

How might we introduce diverse perspectives into ideation to support more reasoned design decisions?



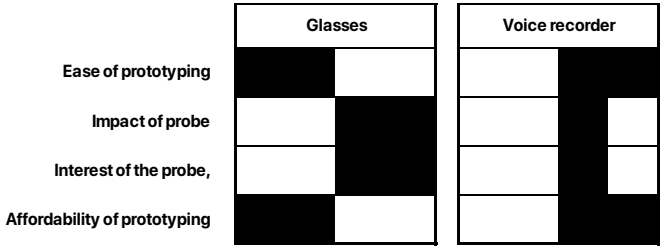


Phase IV: Evaluating

To determine which concept to prototype, each concept was then evaluated using a Harris profile, comparing them across four criteria:

- Impact of the probe
- Interest and richness of the interaction
- Ease of prototyping
- Affordability of prototyping

While the glasses scored highest in conceptual interest, practical constraints like time, complexity, cost, and prototyping difficulty, led to the selection of the voice recorder.



Phase V: Designing Final Concept

In the final phase, the selected concept was translated into a high-fidelity interaction prototype, focusing on how users would experience AI-mediated reflection through a physical device. Focused on interaction logic, system structure, and physical affordances.

System Overview - How it works

The system is designed around a simple interaction: press a button, speak, and receive either structured capture or reflective feedback.

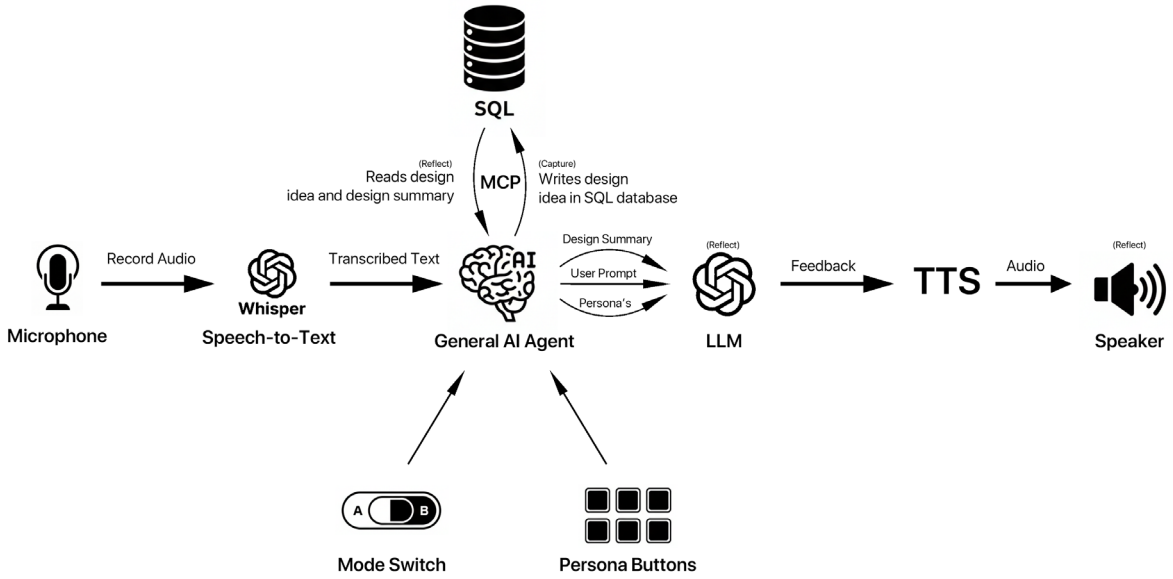
At a high level, the system consists of four layers: input → interpretation → context → response.

1. Voice input
The device features a single record button that activates the microphone. Spoken input is captured and sent to a speech-to-text model (Whisper), converting voice into text.

2. Interpretation and routing
The transcribed text is sent to a general AI agent. Based on the current mode, capture or reflect, the agent determines how the input should be handled.

3. Context management
Using the Model Context Protocol (MCP), the agent reads from or writes to a connected SQL database:

- In capture mode, ideas are structured and stored within the design process.
- In reflect mode, the agent retrieves relevant background information, including a summary of the design process and past ideas, to establish context.



4. Persona-based reflection (Only in Reflect mode)
The system composes a prompt that combines:

- the user's spoken input,
- the retrieved design context (from SQL),
- and the selected persona(s).

A persona-specific AI agent generates reflective feedback, which is then converted to speech using a text-to-speech model and played back through the device speaker.

Physical configuration and components

The prototype is structured as a distributed system. The physical device does not perform on-device processing. Instead, it acts as a controller and audio interface, while all computation is handled by the host system.

- **Physical device**
  Functions as an interface only. It contains buttons, a microphone, a speaker, and a screen, and communicates via Bluetooth (ESP32-S3).
- **Host device**
  Handles all computation and storage. It runs the database, manages communication with AI models and agents, and overlooks interaction logic.

This separation allows the physical device to remain lightweight and minimise latency.

Core components include:

- Microcontroller (processing and communication)
- Microphone (voice input)
- Speaker (audio feedback)
- Record button (press-to-talk interaction)
- Mode switch (capture / reflect)
- Persona selection buttons
- Small display (system state and feedback cues)

Physical Interaction Design

The prototype is designed as a standalone, handheld device with a deliberately minimal set of controls, prioritizing clarity and ease of use. Interaction centers around a single press-to-talk button, enabling quick and intuitive idea capture while providing a clear, tactile affordance.

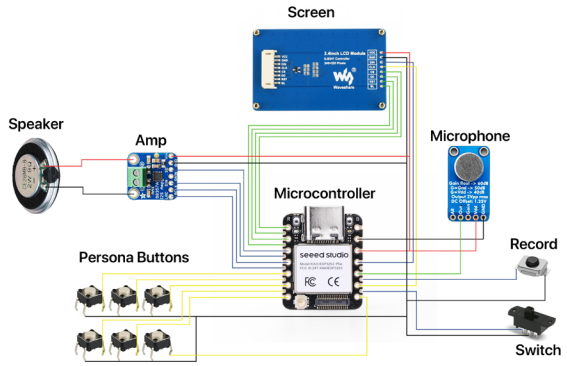
Dedicated persona buttons allow users to enable or disable specific perspectives for AI feedback, with the option to activate multiple personas simultaneously. Physical buttons were intentionally chosen to create a tangible, expressive interaction with the device. In future iterations, these buttons could be replaced by a screen-based interface, enabling users to define custom personas or allowing the AI to predict relevant perspectives based on the design project.

A compact display provides brief visual cues such as system state, battery level, and capture confirmation. The screen also allows users to verify that their spoken input has been correctly captured, offering reassurance.

Ergonomically, the device is sized to support comfortable handheld use or placement on a desk.

A physical model of the device is currently under construction and was not finalized at the time of submission. Documentation of the completed physical prototype will be provided when available.





Component	Purpose / Use
XIAO ESP32-S3 PLUS	Microcontroller; handles logic, button input, and communication with AI services via Bluetooth/Wi-Fi
2.4" SPI TFT Screen	Displays system status, selected personas, mode, and idea capture confirmation
MAX9814 Microphone	Captures user voice for idea input and AI reflection
Dayton Audio CE40-28P-8	Plays back AI responses through text-to-speech
Omron B3F 12×12 mm Buttons	Main tactile buttons for persona selection
Omron B3F 6×6 mm / side button	Secondary or side tactile buttons for mode or additional functions
C&K Miniature Slide Switch	Switch between capture and reflection modes
MAX98357A Class D Amp	Amplifies audio signal to the speaker

Reflection

The public introduction of large language models in late 2022 marked a significant shift in how humans interact with computational systems. It is easy to already take for granted how extraordinary it is that we can communicate with a system that understands human language, reasons about it, and responds in a meaningful way. This project began from that realization: AI introduces an entirely new interaction capability that designers are only beginning to explore.

Abagnale, named after Frank Abagnale Jr., a figure known for convincingly embodying multiple roles, is built around this core capability of AI. The system leverages AI's ability to understand language, reason about intent, and adapt responses to context. Without AI's capacity for personalization and interpretation, this product would not be possible. Unlike traditional software, which is designed to perform a predefined task in a specific way, AI can operate across a wide range of situations once its behavior is shaped. Designing with AI feels less like programming a tool and more like shaping a capability that can be reused in many contexts.

Another key insight from this project is AI's potential for self-agency. Because AI can interpret and respond independently, it can replicate certain human-like roles. In Abagnale, this takes the form of persona-based reflection, where AI acts as a stand-in for different domain experts. This has meaningful implications for design practice. Expertise is expensive, and access to diverse perspectives is often limited, especially for individual designers or small teams. In this sense, Abagnale becomes a design team in your pocket (a deliberate nod to Steve Jobs) offering access to perspectives that would otherwise require significant resources. This can help designers avoid blind spots and make more grounded decisions, even when working alone.

Another problem this project tackles is the one-dimensional tradeoff between capturing ideas and reasoning. Traditionally, stopping to record ideas interrupts deeper thinking, and focusing solely on reasoning risks losing insights. Abagnale solves this: with a single button, designers can capture ideas without breaking flow, turning ideation into an ongoing, dynamic conversation with oneself.

However, the project also surfaced important limitations. One of the most immediate is latency. Human collaborators can respond almost instantly, while AI requires time to reason and generate a response. This delay can make interactions feel unnatural and can disrupt creative flow. While this is currently a technical limitation, it is likely to diminish as AI systems become faster and more optimized.

A second limitation is connectivity. Running AI models on-device is currently impractical for small handheld products, requiring reliance on cloud-based systems. This introduces dependency on internet connectivity, along with associated latency and reliability issues. While future hardware may support on-device hosting, this constraint currently shapes how and where such interactions can occur.

A more fundamental challenge lies in context and ambiguity. AI does not truly understand situations; it infers intent based on provided information. When that context is incomplete or misinterpreted, users can get stuck correcting the system, often creating more confusion. One solution is a simple "retry" or re-reasoning button button that resets the AI to the last prompt, similar to those found in current language model interfaces. This provides users a quick way to recover from misinterpretations.

This led to a further reflection on misinterpretations, specifically in the context of voice-based interaction. Unlike text, speaking naturally encourages users to assume the system possesses human-like understanding, often leading them to under-informing the AI. As a result, the AI may generate responses that are less relevant than those prompted by carefully constructed text. Abagnale addresses this with the idea-capturing feature (this is why this feature exists), its true purpose is actually to give the system the background context it might otherwise miss. This raises an open research question: whether voice-based prompting actually leads to poorer AI responses due to reduced contextual framing.

Ultimately, the central takeaway from this project is that AI's greatest value lies in its capacity to act with self-agency: the ability to interpret, reason, and respond autonomously. Reflecting on the concepts explored during the diverging phase, each concept has relied on this capacity. When a problem requires understanding context and responding by reasoning rather than executing predefined rules, AI becomes a compelling design solution. Designing successful AI interactions, however, requires acknowledging and shaping around its limitations. Particularly latency, context ambiguity, and misinterpretation. When thoughtfully embedded, AI has the potential to enable more personalized, reflective, and natural interactions across future products.

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